BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

BIOLOGY

Past Paper 2022 • Paper Code: Biology Paper–X (01) (22)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** Biology Paper–X (01) (22)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. The largest part of the brain is:

(A) Cerebrum
(B) Thalamus
(C) Cerebellum
(D) Medulla Oblongata

Q2. The union of male and female gametes is called:

(A) Fertilization
(B) Oogenesis
(C) Spermatogenesis
(D) Gametogenesis

Q3. In which disease the walls of the alveoli are broken?

(A) Pneumonia
(B) Bronchitis
(C) Emphysema
(D) Asthma

Q4. Cacti belongs to:

(A) Hydrophytes
(B) Xerophytes
(C) Halophytes
(D) Mesophytes

Q5. Which connective tissue attaches two bones at a joint:

(A) Tendon
(B) Ligaments
(C) Cartilage
(D) Marrow

Q6. Sedative drugs slow down the function of:

(A) Kidney
(B) Lungs
(C) Brain
(D) Stomach

Q7. The function of.....hormone is to help in child birth:

(A) Oxytocin
(B) Progesterone
(C) Estrogen
(D) Thyroxine

Q8. Internal fertilization occurs in:

(A) Fish
(B) Amphibians
(C) Aquatic animals
(D) Mammals

Q9. The gene of interest is isolated from the chromosome of the donor by using:

(A) Endonucleases
(B) Ligases
(C) Calcium ion
(D) Hydrolases

Q10. Which of the given is male gametophyte in flowering plants?

(A) Embryo sac
(B) Stamen
(C) Pollen grain
(D) Megaspore

Q11. An ecological pyramid refers to the:

(A) Pyramid of energy
(B) Pyramid of number
(C) Pyramid of biomass
(D) All these

Q12. The enzymes which are used to cut the DNA in specified sequence is:

(A) Ligase
(B) Protease
(C) DNA ase
(D) Endonuclease

Section-B

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 32

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

QII. Answer any EIGHT parts. Each part carries FOUR marks.

1. Discuss Oogenesis with diagram.
2. How does skin help in homeostasis?
3. What are the biotic components of ecosystem?
4. What is arthritis?
5. Why Mendel selected peas as a subject of studying genetics?

6. Differentiate between epigeal and hypogeal germination.

7. What is breathing? Describe briefly.

8. Differentiate between sensory and motor neurons.

9. State four functions of bones.

10. How is yogurt produced through the use of biotechnology?

11. Why is pituitary gland known as master gland?

Section-C

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 21

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

QIII.

(a) [4] Write a detailed note on Mendel's law of segregation of genes.

(b) [3] What are the causes of air pollution?

QIV.

(a) [4] What are the kidney stones and how they are formed?

(b) [3] Describe the mechanism of breathing.

QV.

(a) [4] Define biogeochemical cycle. Also discuss carbon cycle.

(b) [3] What is genetic engineering?

QVI.

(a) [4] Define cellular respiration. Describe the structure of human respiratory system.

(b) [3] What is dialysis?